

MANUFACTURING PROCESSES – I*Time Allowed: 3 Hours**Full Marks: 70*

Answer to Question No.1 is compulsory and to be answered first.
 This answer is to be made in separate loose script(s) provided for the purpose.
 Maximum time allowed is 45 minutes, after which the loose answer scripts will be collected and
 fresh answer scripts for answering the remaining part of the question will be provided.
 On early submission of answer scripts of Question No.1,
 a student will get the remaining script earlier.

Answer any five questions from the rest.

1. A. Fill in the blanks (any ten): 1x10
- i) The ratio of _____ and acetylene in neutral flame is 1:1.
 - ii) Electrode material for _____ welding is Tungsten.
 - iii) _____ flame is used for welding brass.
 - iv) Axis of the casting and axis of rotation is different in _____ casting process.
 - v) In _____ die-casting method continuous heating of metal is not necessary.
 - vi) _____ allowance is given to compensate effect after solidification.
 - vii) _____ allowance is given for easy removal of pattern.
 - viii) Master pattern is used for making wax pattern for _____ casting method.
 - ix) In _____ welding process arc is not visible to welder.
 - x) Seam welding may be considered as continuous _____ welding.
 - xi) _____ welding is a modification of spot welding.
 - xii) Core box is used to make _____.
 - xiii) Lathe is specified by _____ of the job over carriage.
 - xiv) A small depression in the parent metal during arc welding is called _____.
- B. Choose the correct answer (any ten): 1x10
- i) Solid / split / sweep pattern is made in a single piece.
 - ii) Spot / TIG / Gas welding is an example of resistance welding.
 - iii) Spot / Seam / TIG welding is used in automobile industry.
 - iv) Electrode is not consumed during TIG / MIG / spot welding.
 - v) Projection / Gas / TIG is an example of resistance welding.
 - vi) Spelter is used during soldering / brazing / TIG.
 - vii) Spot welding electrode is made off copper / aluminium / iron.
 - viii) Brazing is similar to soldering / Gas welding / oxy-cutting.
 - ix) Skeleton pattern is used when pattern is big / small.
 - x) Casting is a forming / deforming process.
 - xi) Sanding is a surface finishing / machining process.
 - xii) Brazing is a joining process / removing process.
 - xiii) The filler material used during soldering is called solder / spelter.
 - xiv) Segmental pattern is used for circular / square work.
2. Describe hot chamber and cold chamber die-casting method.
3. Write short notes on – (a) Sweep pattern, (b) Loose-piece pattern. 5+5
4. Describe submerged arc welding process. 5+5

5. What is core print? What is core? Write the advantages of wood as pattern making material. 5+5
 6. How a Lathe machine can be specified? Write the functions of Lathe machine parts? 5+5
 7. Write the difference between up milling and down milling. Write the specifications of drilling machine. 5+5
 8. Write shorts on – (a) Facing, (b) Parting-off, (c) Knurling. 3+3+4
 9. Describe different types of milling cutter. 10
 10. Write short notes on different types of chips. 10
 11. Write the functions of different parts of a radial drilling machine. 10
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